Dr.R.RAJA
Assistant Professor

Contact

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Academic Qualifications: M.Sc., M.Phil., PhD

Teaching Experience: 11 Years

Research Experience: 9 Years

Additional Responsibilities in the University

- Acting as a Warden for the International Student's Hostel from September 2022 to till date.
- Acting as a Compliance Officer to coordinate with FRROs/FROs for ascertaining facts in respect of Foreign Students/ Scholars during grant of various services to them through the FRRO portal.
- Acting as a Coordinator for Students Carrier Progression & Alumni Network (SCPAN) from February 2020 to till date.
- Acting as a Deputy Director for Centre for International Relations (CIR) from August 2018 to till date.
- Acted as a Member of the inspection committee for Revival of Approved Research Centre of Alagappa University affiliated colleges.

- Acted as a Department Research Committee Member for our Affiliated Colleges.
- Acted as a member of the squad team for November 2015 Regular Examinations of our Affiliated Colleges.
- Deputed as a University Representative to oversee the conduct of Distance Education Examination for December 2012 May 2013, May 2014, December 2014, May 2015, December 2015, May 2016 and December 2016, respectively.
- Acted as a member of the organizing committee in the Three-day Orientation Programme to the Fresher's for 2014-2015 batch.

Additional Responsibilities in the Department

- Acted as a Co-ordinator for Village Extension Programme (VEP) for the academic year 2022-2023.
- Acted as a Computer Lab in charge in Department of Mathematics from 2015 to 2018.
- Acted as a Placement Cell coordinator in Department of Mathematics from 2015 to till date.
- Acted as a Student Counselling Cell coordinator in Department of Mathematics from 2015 to till date.
- Member of the Doctoral Committee for the student S. Loganathan (Register No.385) in the Department of Mathematics.
- Act as a member of the squad team for November 2015 Regular Examinations of our Affiliated Colleges.
- Member of the UGC SAP-DRS (Level-I) in the Department of Mathematics.
- Deputed as a University Representative to oversee the conduct of Distance Education Examination for December 2012 May 2013, May 2014, December 2014, May 2015, December 2015, May 2016, December 2016, December 2022, respectively.
- Acted as a member of the organizing committee in the Three-day Orientation Programme to the Fresher's for 2014-2015 batch.
- Acted as a Co-ordinator for Village Placement Programme (VPP) for the academic year 2013-2014.

Areas of Research

Abstract and Fractional Differential Equations, Control Theory, Neural Networks, Stability Analysis of Dynamical Systems, Synchronization and Chaos Theory, Genetic Regulatory and Oscillatory Networks, Multi-Agent Systems, Mathematical Modeling and Population Dynamical Systems, Complex Dynamical Networks.

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	Ph.D.	09	05
	M.Phil.	33	--
Project	PG	95	17
	UG / Others	--	--

Publications

International		National		Others
Journals	Conferences	Journals	Conferences	Books / Chapters / Monographs / Manuals
108	15	--	05	06

Cumulative Impact Factor (as per JCR) : 350.457
h- index : 30
i 10 index : 67
Total Citations : 2383

Funded Research Projects

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	UGC	2013	2015	Studies on Stability Issues of Neural Networks	5.40
2.	AURF	2016	2018	Global Stability Analysis of Neural Networks with Time-varying Delays: The Continuous and Discrete-time Case	3.00
3.	DST-SERB (EMEQ)	2019	2022	Optimal Control of Population Dynamical Systems & Epidemiology: An LMI Approach	14.13

Distinctive Achievements / Awards

1. Awarded “Promising Researcher Award - 2022“ in recognition of the contribution made towards Excellence in Research at Alagappa University.
2. Awarded “Vallal Alagappar Research Recognition Award–2020” in contribution towards the enhancement of Research Outcome of the Alagappa University in the form of "h"-index.
3. Awarded "365 years of Fractional Calculus Award" for the year 2020 in Testimony of High Regard of Achievements in the Area of Fractional Calculus and Its Applications by Biruni University, Turkey.

4. Awarded travel grant for the sum of Rs.1, 80, 000/- from NBHM for attending International Congress of Industrial and Applied Mathematics (ICIAM 2015) in Beijing, China during August 10-14, 2015.
5. Awarded travel grant for the sum of Rs.2, 00, 000/- from NBHM for attending International Congress of Mathematicians (ICM 2014) in Seoul, South Korea during August 15-21, 2014.
6. Sir. C.V. Raman Budding Innovator Award for the year 2010 from Periyar University, Salem.
7. Awarded Senior Research Fellow under Rajiv Gandhi National Fellowship, 2008.
8. Awarded Junior Research Fellow under Rajiv Gandhi National Fellowship, 2005.

Events organized in leading roles

1. Acted as a Co-ordinator for the Village Placement Programme (VPP) for the academic year 2022-2023.
2. Acted as a Convenor for the 2nd International Conference on Mathematical Modeling and Computational Methods in Science and Engineering during January 22-24, 2020.
3. Acted as a Co-ordinator for the Village Exchange Programme (VEP) for the academic year 2018.
4. Acted as a Convenor for the 1st International Conference on Mathematical Modeling and Computational Methods in Science and Engineering during February 20-22, 2017.
5. Acted as a Co-ordinator for the State Level Quiz Contest 2016 on February 15th, 2016.
6. Acted as a Co-ordinator for the Village Placement Programme (VPP) for the academic year 2013-2014.

Events Participated

Conferences / Seminars / Workshops: 15

1. Participated in the *"National Conference on Bridging Innovative Trends in Pure and Applied Mathematics"* held at Bannari Amman Institute of Technology, Erode during July 15-16, 2016.
2. Participated in the *"International Conference on Innovations in Computer Science & Technology (ICICST 2016)"* held at Alagappa University, Karaikudi during April 4-5, 2016.
3. Participated in the *"International Conference on Nonlinear Dynamical Systems (ICNDS 2016)"* held at Bharathiar University, Coimbatore during March 24-26, 2016.
4. Participated in the *"International Congress for Industrial and Applied Mathematics (ICIAM 2015)"*, was held in Beijing, China during August 10 - 14, 2015.
5. Participated in the *"International Congress for Mathematicians (ICM 2014)"*, was held in Seoul, South Korea during August 15 - 21, 2014.
6. Participated in the *"Second National Conference on Advances in Differential Equations and Applications"* held at Periyar University, Salem during March 29-30, 2012.
7. Participated in the National Workshop on *"Decompositions of Graphs and Product Graphs"* held at Periyar University during June 22-25, 2011.

8. Participated in the *"National Conference on Advances in Differential Equations and Applications"* held at Periyar University, Salem during October 7-8, 2010
9. Participated in the *"International Conference on Futuristic Computer Applications"* held at IISC Banglore February 25, 2010.
10. Participated in the *"One day seminar on partial differential equations in commemoration of Prof. E. K. Ramasami's retirement"* held at Bharathiar University, Coimbatore on November 6, 2009.
11. Participated in the *"Fifth Annual Instructional Conference of ADMA and Graph Theory Day V"* held at Periyar University, Salem during June 8-10, 2009.
12. Participated in the National Workshop on *"Mathematical Modeling and Computational Mathematics"* held at Gandhigram Rural University, Gandhigram during February 23-28, 2009.
13. Participated in the *"National Seminar on Algebra and its Applications"* held at Periyar University, Salem during February 12-13, 2009.
14. Participated in the *"15th Ramanujan Symposium on Dynamic Equations"* during February 4-6, 2008.
15. Participated in the *"95th Indian Science Congress held at Andhra University"*, was held in Visakhapatnam during January 3 - 7, 2008.

Other Training Programs

1. Participated in the *"Refresher Course"* organized by the Teaching Learning Centre & Ramanujan College, University of Delhi from March 16-30, 2021.
2. Participated in the *"Refresher Course"* organized by the Academic Staff College, Pondicherry University, Puducherry from September 7-27, 2016.
3. Participated in the *"Orientation Programme"* organized by the Academic Staff College, Madurai Kamaraj University, Madurai from May 30-June 29, 2013.
4. Participated in the *"Orientation Programme"* organized by the Curriculum Development Cell, Alagappa University, Karaikudi from August 24-31, 2012.

Overseas Exposure / Visits

1. Visited Rome, Italy and presented a research paper in the 3rd International Conference and Summer School on Numerical Computations: Theory and Algorithms during June14-20, 2019.
2. Visited Maejo University, Chiang Mai, Thailand as a Visiting Faculty during 30th August to 25th September 2018.
3. Visited Beijing, China for attending the "*International Congress of Industrial and Applied Mathematics* (ICIAM 2015)" in during August 10-14, 2015.
4. Visited Seoul, South Korea for attending the "*International Congress of Mathematicians (ICM 2014)"* in during August 15-21, 2014.

Membership in

Editorial Board

1. Chief Editor for the Conference Proceedings "2nd ICMMCMSE-2020" in the IOP Journal of Physics: Conference Series (Institute of Physics Publishers).
2. Guest Editor for the special issue on some selected high-quality papers from the "2nd ICMMCMSE-2020" in Mathematics and Computers in Simulation (Elsevier Publication).
3. Guest Editor for the special issue on some selected high-quality papers from the "ICMA-MU (2018)" in Mathematics and Computers in Simulation (Elsevier Publication).
4. Guest Editor for the special issue "Mathematical Problems and Computational Methods in Science and Engineering" in the International Journal of Advanced Intelligence Paradigms. (Inderscience Enterprises Ltd).
5. Guest Editor for the special issue "Computational Problems in Science and Engineering" in the International Journal of Computational Systems in Engineering (Inderscience Enterprises Ltd).
6. Guest Editor for the special issue on some selected high-quality papers from the "ICMMCMSE-2017" in Mathematics and Computers in Simulation (Elsevier Publication).
7. Guest Editor for the special issue *"Dynamics of Neural Networks and Applications"* in Optimization in Mathematical Problems in Engineering (Hindawi Publishers).
8. Guest Editor for the special issue *"Stability and Control of Stochastic Systems"* in Mathematical Problems in Engineering (Hindawi Publishers).
9. Guest Editor for the special issue *"Stability and Its Application of Stochastic Systems"* in the International Journal of Differential Equations (Hindawi Publishers).
10. Editor for the *"DJ Journal of Engineering and Applied Mathematics"* from 2015 to till date editorial board for the *"World Academy of Science, Engineering and Technology"* from 2016 to till date.

Advisory Board

1. Member of Board of Studies in the School of Mathematics, Alagappa University, Karaikudi from 2012 to till date.
2. Member of the International Scientific Committee in *2016 International Conference on Mathematical, Computational and Statistical Sciences and Engineering (MCSSE2016)* which will be held on October 30-31, 2016 in Shenzhen, China.
3. Member of the Technical Programme Committee in *2016 International Conference on Mechanical Design and Control Engineering [MDCE2016]* which will be held on November 19-20, 2016 in Wuhan, China.

Others

1. Member of the Reviewer board in Advances in Difference Equations.
2. Member of the Reviewer board in American Mathematical Society.
3. Member of the Reviewer board in Canadian Journal of Physics.
4. Member of the Reviewer board in Circuits, Systems and Signal Processing.
5. Member of the Reviewer board in Computers and Mathematics with Applications.
6. Member of the Reviewer board in Communications in Computer and Information Science.

7. Member of the Reviewer board in IEEE Transactions on Neural Networks.
8. Member of the Reviewer board in Intelligent Control and Automation.
9. Member of the Reviewer board in International Journal of Machine Learning and Cybernetics.
10. Member of the Reviewer board in Journal of The Franklin Institute.
11. Member of the Reviewer board in Journal of Applied Mathematics and Mechanics.
12. Member of the Reviewer board in Mathematical Problems in Engineering.
13. Member of the Reviewer board in Neural Computing and Applications.
14. Member of the Reviewer board in Neurocomputing.
15. Member of the Reviewer board in Neural Networks.
16. Member of the Reviewer board in Nonlinear Analysis: Hybrid Systems.
17. Member of the Reviewer board in International Journal of Advanced Computer Science and Applications.
18. Member of the Reviewer board in Nonlinear Dynamics.
19. Member of the Reviewer board in IEEE Transactions on Cybernetics.
20. Member of the Reviewer board in Applied Mathematics and Computation.
21. Member of the Reviewer board in International Academy of Computer \& IT Engineering.
22. Member of the Reviewer board in Mathematics and Computers in Simulation.
23. Member of the Reviewer board in International Journal of Robust & Nonlinear Control.
24. Member of the Reviewer board in International Journal of System Science.
25. Member of the Reviewer board in IEEE Transactions on Cybernetics.
26. Member of the Reviewer board in Mathematics MDPI.
27. Member of the Reviewer board in AIMS Mathematics.
28. Member of the Reviewer board in Chaos, Solitons and Fractals.
29. Member of the Reviewer board in Asian Journal of Control.
30. Member of the Reviewer board in Neural Processing Letters.
31. Member of the Reviewer board in Neural Computing and Applications.
32. Member of the Reviewer board in Fractal and Fractional.
33. Member of the Reviewer board in Mathematical Methods in the Applied Sciences.
34. Member of the Reviewer board in International Journal of Adaptive Control & Signal Processing.

Resource persons in various capacities

1. Invited as a **Chief Guest** and delivered a talk in the *"PG Mathematics Association"*, Department of Mathematics, Seethalakshmi Achi College for Women, Pallathur on September 3, 2016.
2. Delivered a **talk** in the *"National Conference on Recent Trends in Applied Mathematics (NCRTAM 16)"* organized by PG and Research Department of Mathematics, St. Joseph's College of Arts and Science (Autonomous), Cuddalore during July 22-23, 2016.
3. Delivered a **lecture** in the 20 day *"Young Scientist Programme"* organized by Periyar University, Salem and Tamilnadu State Council for Science and Technology, Chennai from May 02-21, 2011.

4. Invited as **keynote speaker** in International Workshop on Fractional Order Systems: Theory and Applications held by Aurel Vlaicu University of Arad, Romania and delivered a talk about "Mathematical Modeling in Artificial Intelligence" on 29 November 2021.
5. Delivered as **invited speaker** in International Conference on Mathematics, Statistics and Data Science (ICMSDS-2022) held by Mahatma Gandhi University Kottayam, India, in which he talked about "Mathematical Modeling and its Applications to Neural Networks" on 26-27 February 2022.
6. Invited as **keynote speaker** in One Day International Conference on Analytic Hierarchy Process and Artificial Intelligence held by Kandaswami Kandar's College, Velur, Tamilnadu, India on 5 August 2022.
7. Invited as a **Chief Guest** in the International Research Outreach Programme (IROP-2022) held in hybrid mode on September 16, 2022, Organized by Faculty of Mathematics- Informatics Teacher Education, Dong Thap University, Vietnam.
8. Invited as **keynote speaker** in International Seminar/ Webinar on Neoteric Applications in Mathematics conducted by Thiagarajar College, Madurai in collaboration with school of mathematics, alagappa University, Karaikudi and delivered a talk about "Why modelling is necessary for real life situation?" during 8th - 9th November 2022.

Others

1. Articles published in Newspapers / Magazines: xx
2. Products developed: xx
3. No. of PhD Thesis evaluated : **01**
4. No. of PhD Public Viva Voce Examination conducted: **09**
5. Sequences submitted in Gen Bank: xx

**Social Interests and Initiatives / Articles in Newspapers etc can also be included

Recent Publications

2023

- Sayooj Aby Jose, **R. Raja**, B. I. Omede Ravi P. Agarwal, J. Alzabut, J. Cao, V. E. Balas, Mathematical modeling on co-infection: transmission dynamics of Zika virus and Dengue fever, ***Nonlinear Dynamics,*** Vol.111, Iss.5, Pp. 4879-4914. **(IF:5.741)**.
- Sayooj Aby Jose, **R Raja**, J Dianavinnarasi, D Baleanu, A Jirawattanapanit, Mathematical modeling of chickenpox in Phuket: Efficacy of precautionary measures and bifurcation analysis, ***Biomedical Signal Processing and Control***, Vol.84, **(IF: 5.076)**.
- Dianavinnarasi Joseph, **Raja Ramachandran**, Jehad Alzabut, Sayooj Aby Jose, Hasib Khan, A Fractional-Order Density-Dependent Mathematical Model to Find the Better Strain of Wolbachia, ***Symmetry***, vol. 15, Iss.4, **(IF: 2.940)**

2022

- S.A. Jose, **R. Raja**, Q. Zhu, J. Alzabut, M. Niezabitowski, V.E. Balas, (2022), "An Integrated Eco-Epidemiological Plant Pest Natural Enemy Differential Equation Model with Various Impulsive Strategies", ***Mathematical Problems in Engineering***, Vol.2022, Article ID 4780680, Pp.1-23. **(IF: 1.430).**
- Sayooj Aby Jose**, Raja Ramachandran**, Dumitru Baleanu, Hasan S. Panigoro, Jehad Alzabut, Valentina E. Balas, Computational dynamics of a fractional order substance addictions transfer model with Atangana-Baleanu-Caputo derivative, ***Mathematical methods in the applied sciences,*** vol. 46, iss.5, Pp. 5060-5085. **(IF: 3.007)**.
- Sayooj Aby Jose, R. Raja, J. Cao, J. Alzabut, M. Niezabitowski, Valentina E. Balas, Stability Analysis and Comparative Study on different Eco-epidemiological Models: Stage Structure for Prey and Predator Concerning Impulsive Control, ***Optimal Control, Applications and Methods***, (43)(3), 842-866, 2022. **(IF: 1.955)**
- A. Pratap, **R. Raja**, J. Cao, C. Huang, J. Alzabut, O. Bagdasar, (2022), "O(t^{-β}) O (t − β) Synchronization and Asymptotic Synchronization of Delayed Fractional Order Neural Networks", ***Acta Mathematica Scientia***, Vol. 42, Iss.4, Pp.1273-1292, **(IF: 1.085)**
- S. Aadhithiyan, **R. Raja**, Q. Zhu, J. Alzabut, M. Niezabitowski, C.P. Lim, (2022), "A Robust Non-Fragile Control Lag Synchronization for Fractional Order Multi-Weighted Complex Dynamic Networks with Coupling Delays", ***Neural Processing Letters***, Pp.1-22. **(IF: 2.565)**
- S.A. Jose, **R. Raja**, J. Alzabut, G. Rajchakit, J. Cao, V.E. Balas, (2022), "Mathematical modeling on transmission and optimal control strategies of corruption dynamics", ***Nonlinear Dynamics***, Vol.109, Iss.4, Pp.3169-3187. **(IF: 5.741)**
- S.A. Jose, **R. Raja**, Q. Zhu, J. Alzabut, M. Niezabitowski, V.E. Balas, (2022), "Impact of strong determination and awareness on substance addictions: A mathematical modeling approach", ***Mathematical Methods in the Applied Sciences***, Vol.45, Iss.8, Pp.4140-4160. **(IF: 3.007)**
- A. Stephen, **R. Raja**, J. Alzabut, Q. Zhu, M. Niezabitowski, O. Bagdasar, (2022), "Mixed time delayed nonlinear multi-agent dynamic systems for asymptotic stability and non-fragile synchronization criteria", ***Neural Processing Letters***, Vol.54, Iss.1, Pp.43-74. **(IF: 2.565)**
- S. Aadhithiyan, **R. Raja**, J. Alzabut, Q. Zhu, M. Niezabitowski, (2022), "Robust non-fragile Mittag-Leffler synchronization of fractional order non-linear complex dynamical networks with constant and infinite distributed delays", ***Mathematical Methods in the Applied Sciences***, Vol.45, Iss.4, Pp.2166-2189. **(IF: 3.007)**
- J. Dianavinnarasi, **R. Raja**, J. Alzabut, J. Cao, M. Niezabitowski, O. Bagdasar, (2022), "Application of Caputo–Fabrizio operator to suppress the Aedes Aegypti mosquitoes via Wolbachia: An LMI approach", ***Mathematics and Computers in Simulation***, Vol. 201, Pp.462-485. **(IF: 3.601)**
- A. Pratap, **R. Raja**, R.P. Agarwal, J. Alzabut, M. Niezabitowski, E. Hincal, (2022), "Further results on asymptotic and finite-time stability analysis of fractional-order time-delayed genetic regulatory networks", ***Neurocomputing***, Vol.475, Pp.26-37, **(IF: 5.779)**
- M. Iswarya, **R. Raja**, J. Cao, M. Niezabitowski, J. Alzabut, C. Maharajan, (2022), "New results on exponential input-to-state stability analysis of memristor based complex-valued

inertial neural networks with proportional and distributed delays", ***Mathematics and Computers in Simulation***, Vol.201, Pp.440-461, **(IF: 3.601)**

- R. Thomas, SA. Jose, **R. Raja**, J. Cao, V.E. Balas, (2022), "Modeling and analysis of SEIRS epidemic models using homotopy perturbation method: A special outlook to 2019-nCoV in India", ***International Journal of Biomathematics***, Vol.15, Iss.8, ID:2250059. **(IF: 2.129)**
- S.O. Neill, O. Bagdasar, S. Berry, N. Popovici, **R. Raja**, (2022), "Modelling equilibrium for a multi-criteria selfish routing network equilibrium flow problem", ***Mathematics and Computers in Simulation***, Vol.201, Pp. 658–669. **(IF: 3.601)**
- M. Iswarya, **R. Raja**, J. Cao, J. Alzabut, C. Maharajan, (2022), "New results on exponential input-to-state stability analysis of memristor based complex-valued inertial neural networks with proportional and distributed delays", ***Mathematics and Computers in Simulation***, Vol.201, Pp. 440–461. **(IF: 3.601)**
- J. Dianavinnarasi, **R. Raja**, J. Alzabut, M. Niezabitowski, O. Bagdasar, (2022), "Application of Caputo–Fabrizio operator to suppress the Aedes Aegypti mosquitoes via Wolbachia: An LMI approach", **Mathematics *and Computers in Simulation***, Vol.201, Pp. 462–485. **(IF: 3.601)**

2021

- J. Dianavinnarasi, **R. Raja,** J. Alzabut, M. Niezabitowski, G. Selvam, (2021), "An LMI Approach-Based Mathematical Model to Control Aedes aegypti Mosquitoes Population via Biological Control", ***Mathematical Problems in Engineering***, Vol. 2021, Article ID 5565949, Pp.1-18. **(IF: 1.430)**
- J. Dianavinnarasi, **R. Raja**, J. Alzabut, M. Niezabitowski, O. Bagdasar, (2021), "Controlling Wolbachia transmission and invasion dynamics among aedes aegypti population via impulsive control strategy", ***Symmetry***, Vol.13, Iss.3, ID: 434. **(IF: 2.940)**
- M. Iswarya, **R. Raja**, Q. Zhu, M. Niezabitowski, J. Alzabut, C. Maharajan, (2021), "Existence, uniqueness, and exponential stability of uncertain delayed neural networks with inertial term: nonreduced order case", ***Mathematical Problems in Engineering***, 2021, Article ID 5560763, Pp.1-15. **(IF: 1.430)**
- S. Senthilraj, T. Saravanakumar, **R. Raja**, J. Alzabut, (2021), "Delay-dependent passivity analysis of nondeterministic genetic regulatory networks with leakage and distributed delays against impulsive perturbations", ***Advances in Difference Equations***, Vol.2021, Iss.1, Pp.1-26. **(IF: 3.702)**
- A. Stephen, **R. Raja**, J. Alzabut, Q. Zhu, M. Niezabitowski, C.P. Lim, (2021), "A Lyapunov– Krasovskii functional approach to stability and linear feedback synchronization control for nonlinear multi-agent systems with mixed time delays", ***Mathematical Problems in Engineering***, Vol.2021 **(IF: 1.430)**
- S. Aadhithiyan, **R. Raja**, Q. Zhu, J. Alzabut, M. Niezabitowski, C.P. Lim, (2021), "Modified projective synchronization of distributive fractional order complex dynamic networks with model uncertainty via adaptive control", ***Chaos, Solitons & Fractals***, Vol.147, ID: 110853. **(IF: 9.922)**
- S. Aadhithiyan, **R. Raja**, Q. Zhu, J. Alzabut, M. Niezabitowski, C.P. Lim, (2021), "Exponential synchronization of nonlinear multi-weighted complex dynamic networks

with hybrid time varying delays", ***Neural Processing Letters***, Vol. 53, Iss.2, Pp.1035-1063. **(IF: 2.565)**

- A. Pratap, **R. Raja**, J. Cao, C. Huang, M. Niezabitowski, O. Bagdasar, (2021), "Stability of discrete-time fractional-order time-delayed neural networks in complex field", ***Mathematical Methods in the Applied Sciences***, Vol. 44, Iss.1, Pp.419-440. **(IF: 3.007)**

2020

- B.S. Vadivoo, **R. Raja**, J. Cao, G. Rajchakit, A.R. Seadawy, (2020), "Controllability criteria of fractional differential dynamical systems with non-instantaneous impulses", ***IMA Journal of Mathematical Control and Information***, Vol. 37, Iss.3, Pp.777-793. **(IF: 1.48)**
- C. Sowmiya, **R. Raja**, J. Cao, G. Rajchakit, A. Alsaedi, (2020), "A delay-dependent asymptotic stability criteria for uncertain BAM neural networks with leakage and discrete time-varying delays: A novel summation inequality", ***Asian Journal of Control***, Vol.22, Iss.5, Pp.1880-1891. **(IF: 2.444)**
- J. Dianavinnarasi, Y. Cao, **R. Raja**, G. Rajchakit, C.P. Lim, (2020), "Delay-dependent stability criteria of delayed positive systems with uncertain control inputs: Application in mosquito-borne morbidities control", ***Applied Mathematics and Computation***, Vol.382, ID: 125210. **(IF: 4.397)**
- T. Saravanakumar, V.J. Nirmala, **R. Raja**, J. Cao, G. Lu, (2020), "Finite-time reliable dissipative control of neutral-type switched artificial neural networks with non-linear fault inputs and randomly occurring uncertainties", ***Asian Journal of Control***, Vol.22, Iss.6, Pp.2487-2499. **(IF: 2.444)**
- A. Pratap, **R. Raja**, R. Agarwal, J. Cao, O. Bagdasar, (2020), "Multi-weighted complex structure on fractional order coupled neural networks with linear coupling delay: a robust synchronization problem", ***Neural Processing Letters***, Vol. 51, Iss.3, Pp.2453-2479. **(IF: 2.565)**
- A. Pratap, **R. Raja**, J. Cao, J. Alzabut, C. Huang, (2020), "Finite-time synchronization criterion of graph theory perspective fractional-order coupled discontinuous neural networks", ***Advances in Difference Equations***, Vol. 2020, Iss.1, Pp.1-24. **(IF: 3.702)**
- A. Pratap, **R. Raja**, C. Sowmiya, O. Bagdasar, J. Cao, G. Rajchakit, (2020), "Global projective lag synchronization of fractional order memristor based BAM neural networks with mixed time varying delays", ***Asian Journal of Control***, Vol.22, Iss.1, Pp.570-583. **(IF: 2.444)**
- A. Pratap, **R. Raja**, J. Cao, F.A. Rihan, A.R. Seadawy, (2020), "Quasi-pinning synchronization and stabilization of fractional order BAM neural networks with delays and discontinuous neuron activations", ***Chaos, Solitons & Fractals***, Vol.131, ID:109491. **(IF: 9.922)**
- G. Rajchakit, P. Chanthorn, M. Niezabitowski, **R. Raja**, D. Baleanu, A. Pratap, (2020), "Impulsive effects on stability and passivity analysis of memristor-based fractional-order competitive neural networks", ***Neurocomputing,*** Vol.417, Pp.290-301. **(IF: 5.779)**
- A. Pratap, **R. Raja**, J. Alzabut, J. Dianavinnarasi, J. Cao, G. Rajchakit, (2020), "Finite-time Mittag Leffler stability of fractional-order quaternion-valued Memristive neural

networks with impulses", ***Neural Processing Letters***, Vol. 51, Iss.2, Pp.1485-1526. **(IF: 2.565)**

- A. Pratap, **R. Raja**, J. Alzabut, J. Cao, G. Rajchakit, C. Huang, (2020), "Mittag-Leffler stability and adaptive impulsive synchronization of fractional order neural networks in quaternion field", ***Mathematical Methods in the Applied Sciences,*** Vol.43 Iss.10, Pp.62236253. **(IF: 3.007)**

2019

- O. Bagdasar, S. Berry, S.O.Neill, N. Popovici, **R. Raja**, (2019), "Traffic assignment: Methods and simulations for an alternative formulation of the fixed demand problem", ***Mathematics and Computers in Simulation***, Vol.155, Pp.360-373, **(IF: 3.601)**
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